

LESIA SEMENOVA

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My research develops the foundations, algorithms, applications, and practice of multiplicity-aware trustworthy AI. Rather than treating learning as the search for a single best model, I study the many models and representations that can perform approximately equally well on the same dataset and ask what this multiplicity reveals about reliability, interpretability, robustness, and privacy. My work has established when simpler-yet-accurate models can exist, how model multiplicity changes robust algorithmic recourse, how deploying multiple models can affect privacy-robustness trade-offs, and how multiplicity can support safer decision-making in domains such as healthcare and public-sector data science. I also build community around this agenda through workshops and tutorials, conference leadership, and mentoring programs spanning undergraduate, graduate, and international researchers. My goal is to make multiplicity a practical foundation for AI systems whose uncertainty can be understood, measured, stress-tested, and governed.

RESEARCH INTERESTS

Safe and Trustworthy AI, Interpretability, Uncertainty, Model Multiplicity, Machine Learning,
Human-Centered Design, AI in Healthcare, Reinforcement Learning, Reasoning

EMPLOYMENT HISTORY

Assistant Professor, Rutgers University, New Brunswick, NJ, USA July 2025 – Present
Department of Computer Science

Postdoctoral Researcher, Microsoft Research, NYC, USA July 2024 – June 2025
Machine Learning Team

Research Intern, Pinterest Labs, Palo Alto, CA, USA Summer 2020, Summer 2021
Applied Science Team

Software Engineer, Samsung R&D Institute, Kyiv, Ukraine 2012 – 2014
Interaction Lab, Augmented Reality Team

EDUCATION

Duke University, Durham, NC, USA
Ph.D. in Computer Science 2016 – 2024
Advisors: Cynthia Rudin, Ronald Parr

Taras Shevchenko National University of Kyiv, Kyiv, Ukraine
MS in Applied Mathematics 2012 – 2014
BS in Applied Mathematics 2008 – 2012

PUBLICATIONS

[Google Scholar](#)

>2.3k citations

(* denotes equal
contributions)

- 21 Ethan Hsu, Harry Chen, Chudi Zhong, **Lesia Semenova**. The Double-Edged Nature of the Rashomon Set for Trustworthy Machine Learning. *International Conference on Machine Learning (ICML)*, 2026
(**Spotlight, 2.2%**).
- 20 Michal Moshkovitz*, Suraj Srinivas*, **Lesia Semenova***, Nave Frost, Cyrus Rashtchian, Valentyn Boreiko, Shichang Zhang, Himabindu Lakkaraju, Cynthia Rudin, Jennifer Wortman Vaughan. Position: Explainability Research Must Prioritize Foundations over Ad-hoc Methods. *International Conference on Machine Learning (ICML)*, 2026.
- 19 Shihan Feng, Cheng Zhang, Michael Xi, Ethan Hsu, **Lesia Semenova**, Chudi Zhong. Parameter-Efficient Construction of the Rashomon Slice for Concept Bottleneck Models. *Transactions on Machine Learning Research*, 2026.
- 18 Harry Chen, Yiyang Sun, Michal Moshkovitz, Zachery Boner, **Lesia Semenova**, Cynthia Rudin, Ronald Parr. Noise as a Natural Regularizer in Markov Decision Processes: Connecting Environmental Stochasticity and Policy Simplicity. *International Conference on Machine Learning (ICML)*, 2026.
— An **oral workshop version** appeared at *Workshop on Interpretable Policies in Reinforcement Learning @RLC-2024*

- 17 Yingfan Wang, German G Gornalusse, Urja Bhatt, David A Siegel, Alton Barbehenn, Rebecca Hoh, Jeffrey Martin, Frederick Hecht, Christopher Pilcher, **Lesia Semenova**, David M Murdoch, David M Margolis, Claire N Levy, Keith R Jerome, Cynthia D Rudin, Florian Hladik, Steven G Deeks, Sulggi A Lee, Edward P Browne. Distinct Transcriptomic Profiles in ART-Treated People with HIV Are Associated with NF- κ B-regulated Gene Expression and Inflammatory Cytokine Signatures. *JAIDS Journal of Acquired Immune Deficiency Syndromes*, 2026.
- 16 Bohdan Turbal, Iryna Voitsitska, **Lesia Semenova**. ElliCE: Efficient and provably robust algorithmic recourse via the Rashomon sets. *Neural Information Processing Systems (NeurIPS)*, 2025 (**Spotlight, 3.2%**).
- 15 Ethan Hsu, Tony Cao, **Lesia Semenova**, Chudi Zhong. The Rashomon Set Has It All: Analyzing Trustworthiness of Trees under Multiplicity. *Neural Information Processing Systems (NeurIPS)*, 2025.
- 14 Dennis Tang, Jon Donnelly, Alina Jade Barnett, **Lesia Semenova**, Jin Jing, Peter Hadar, Ioannis Karakis, Olga Selioutski, Kehan Zhao, M. Brandon Westover, Cynthia Rudin. This EEG looks like these EEGs: Interpretable interictal epileptiform discharge detection with ProtoEEG-kNN. *International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI)*, 2025.
— An [oral workshop version](#) appeared at 2023 *Workshop on Medical Imaging meets NeurIPS*
- 13 Gaurav Rajesh Parikh, Jenny Huang, Albert Sun, **Lesia Semenova**, Cynthia Rudin. Navigating progress: Enhancing public transit for more equitable communities via interpretable causal inference. *Harvard Data Science Review*, 2025.
— A [workshop version](#) appeared at *NeurIPS 2022 Workshop on Causality for Real-world Impact*
— **Won 2022 American Statistical Association Data Challenge Expo Student Competition**
- 12 Chloe Qinyu Zhu, Muhang Tian, **Lesia Semenova**, Jiachang Liu, Jack Xu, Joseph Scarpa, Cynthia Rudin. Fast and Interpretable Mortality Risk Scores for Critical Care Patients. *Journal of the American Medical Informatics Association (JAMIA)*, 2025.
- 11 Zachery Boner*, Harry Chen*, **Lesia Semenova***, Ronald Parr, Cynthia Rudin. Using noise to infer aspects of simplicity without learning. *Neural Information Processing Systems (NeurIPS)*, 2024.
- 10 Cynthia Rudin, Chudi Zhong, **Lesia Semenova**, Margo Seltzer, Ronald Parr, Jiachang Liu, Srikar Katta, Jon Donnelly, Harry Chen, Zachery Boner. Amazing things come from having many good models. *International Conference on Machine Learning (ICML)*, 2024 (**Spotlight, 3.85%**).
- 9 Siong Thye Goh*, **Lesia Semenova***, Cynthia Rudin. Sparse density trees and lists: an interpretable alternative to high-dimensional histograms. *INFORMS Journal on Data Science (IJDS)*, 2024.
- 8 **Lesia Semenova**, Yingfan Wang, Shane Falcinelli, Nancie Archin, Alicia D Cooper-Volkheimer, David M Margolis, Nilu Goonetilleke, David M Murdoch, Cynthia D Rudin, Edward P Browne. Machine learning approaches identify immunologic signatures of total and intact HIV DNA during long-term antiretroviral therapy. *eLIFE*, 2024
- 7 **Lesia Semenova**, Harry Chen, Ronald Parr, Cynthia Rudin. A path to simpler models starts with noise. *Neural Information Processing Systems (NeurIPS)*, 2023.
- 6 **Lesia Semenova**, Cynthia Rudin, Ronald Parr. On the existence of simpler machine learning models. *ACM Conference on Fairness, Accountability, and Transparency (FAccT)*, 2022.
- 5 Cynthia Rudin, Chaofan Chen, Zhi Chen, Haiyang Huang, **Lesia Semenova**, Chudi Zhong. Interpretable machine learning: fundamental principles and 10 grand challenges. *Statistics Surveys*, 2022.
- WORKSHOPS AND PREPRINTS 4 Alex Oesterling, Angikar Ghosal, Haoyang Yu, Rui Xin, Yasa Baig, **Lesia Semenova**, Cynthia Rudin. Multitask learning for citation purpose classification. *Second Workshop on Scholarly Document Processing, ACL*, 2021.
— **Won third place in 3C Shared Task Competition**
- 3 Panyu Chen, Yiyang Sun, Zachery Boner, **Lesia Semenova**, Cynthia Rudin. When Are Simple Models Statistically Indistinguishable From the Best Possible Models? Under review, 2026.

	2	Neil Gandhi, Maria George, Harry Chen, Lesia Semenova . RECLAIM: Robust Algorithmic Recourse with Actionable Missingness. Under review, 2026.	
	1	Wu, Yuan, Mairui Li, Lesia Semenova , and Chudi Zhong. The Asymmetric Harms of LLM Compression. arXiv:2608.19670, 2026.	
INVITED TALKS		ASA Section on Statistical Learning and Data Science Conference (incoming)	2026
		2nd NJIT Workshop on Multimedia Intelligence	2026
		ELLIS Workshop on Safe and Secure Artificial Intelligence	2026
		Rutgers, Signal and Information Processing Seminar Series	2026
		University of California San Diego EnCORE Workshop on Interpretability in Modern AI	2026
		University of Rhode Island	2026
		Rutgers, New Research Frontiers in Education, Technology, and Society	2025
		Rutgers, Undergraduate Student Alliance of Computer Scientists	2025
		Joint Statistics Meetings (JSM)	2025
		18th International Joint Conference on Computational and Financial Econometrics (CFE) and Computational and Methodological Statistics (CMStatistics)	2024
		INFORMS Annual Meeting	2024
		Theory of Interpretable AI Seminar	2024
		The 25th International Symposium on Mathematical Programming	2024
		Johns Hopkins University, Applied Physics Lab, AI/ML seminar	2024
		Oden Institute, UT Austin	2024
		Lawrence Livermore National Laboratory	2024
		Microsoft Research	2024
		Conference on Information Sciences and Systems (CISS)	2023
		Joint Statistics Meetings (JSM), Near-Optimization Topic-Contributed session	2021
		INFORMS Annual Meeting	2020
SELECTED AWARDS AND HONORS		Silver Reviewer at ICML	2026
		Top Reviewer at ICML	2025
		PhD Dissertation Award from the Department of Computer Science at Duke University	2024
		Rising Stars in Computational and Data Science, University of Texas at Austin	2024
		1st place in American Statistical Association Data Challenge Expo Student Competition	2023
		1st place in American Statistical Association Data Challenge Expo Student Competition	2022
		3rd place in 3C Shared Task Competition	2021
		SAMSI Fellowship	2018
		TA Award Honorable Mention	2018
		Duke CS Department Fellowship	2016 – 2018
		KDD travel award	2016
		Scholarship of Mayor of Kyiv	2012 – 2014
		Scholarship of Zavtra.UA competition	2012-2013 and 2013 – 2014
		Scholarship of Academic Council of Taras Shevchenko National University	2011 – 2012
		Scholarship of President of Ukraine	2008 – 2009
ADVISING AND MENTORING		Zhipeng Yang, PhD Student, Rutgers University	2026-present
		Neil Gandhi, undergraduate, Rutgers University	2025-present
		Demian Samus, master, Taras Shevchenko National University of Kyiv	2026-present
		Zihua Meng, master, University of Pennsylvania	2026-present
		Andrew Chung, master, Columbia University	2026-present
		Timothy Hsu, undergraduate, Rutgers University (Aresty)	2026-present
		Rutva Patel, master, University of Pennsylvania	2026-present
		Nikhil Sethuraman, undergraduate, Rutgers University (Aresty)	2026-present
		Aryaman Mishra, undergraduate, Rutgers University (Aresty)	2026-present
		Maria George, undergraduate, Rutgers University	2026-present
		Abhishu Oza, master, Rutgers University	2026-present

Prisha Priyadarshini, undergraduate, Rutgers University 2026-present

Former Advisees

Arnav Sirigere, undergraduate, UIUC (DIMACS REU)	Summer 2026
Michael Xi, PhD Student, Rutgers University	2025-2026
Anastasia Mazur, master, Ukrainian Catholic University (RAI4Ukraine)	2025-2026
Ostap Pavlyshyn, undergraduate, Ukrainian Catholic University (RAI4Ukraine)	2025-2026
Bohdan Turbal, undergraduate, Taras Shevchenko National University of Kyiv (RAI4Ukraine, now PhD at Princeton)	2024-2026
Shihan Feng, master, Duke University (now PhD student at UNC)	2024-2026
Iryna Voitsitska, undergraduate, Ukrainian Catholic University (RAI4Ukraine)	2024-2026
Ethan Hsu, undergraduate, Duke University (now at Amazon)	2024-2026
Zach Boner, PhD Student, Duke University	2023-2024
Dennis Tang, undergraduate, Duke University (now PhD student at Duke)	2023-2025
Chloe Zhu, undergraduate, Duke University (now PhD student at Duke)	2023-2024
Muhang (Tony) Tian, undergraduate, Duke University (now PhD student at NYU)	2023-2024
Harry Chen, undergraduate, Duke University (now PhD student at MIT)	2022-2026
Flora Shi, undergraduate, Duke University (now PhD student at MIT)	2022-2023

Co-instructor for Duke Data Science teams that participated in various Data Science competitions
(9 teams total, 29 Duke undergraduate students)

TEACHING

Rutgers University

- CS536 Machine Learning II	Fall 2026
- CS671 Interpretable and Explainable AI	Spring 2026

Certificate in College Teaching, Duke Graduate School

Formal pedagogical training in the college teaching

Institute of Advanced Study

2022 Program for women and mathematics: "[The Mathematics of Machine Learning](#)"

Terng Lecture Course on Interpretable Machine Learning May 2022

SERVICE

Workshop Organizer:

- [Navigating Model Uncertainty and the Rashomon Effect: From Theory and Tools to Applications and Impact Workshop @ AAI 2026](#)
- [Interpretable AI: Past, Present and Future Workshop @ NeurIPS 2024](#)

Tutorial Organizer and Presenter:

- Translation Tutorial: [The Illusion of the Best Model —Multiplicity, Interpretability, and Accountability in High-Stakes AI @ FAccT 2026](#)
- [From Underspecification to Alignment: Breaking the One-Model Mindset for Reliable AI @AAAI 2026](#)

Mentor:

- Doctoral Colloquium @ ACM FAccT	2026
- Women in Machine Learning Workshop @ NeurIPS	2024, 2025
- "Artificial Intelligence for Art and Fun" capstone event as a part of Duke's FEMMES+ (Females and Allies Excelling More in Math, Engineering, and Science) outreach program to introduce young female students (4 th -6 th grade) to math, science, and engineering	2021

Research advisor:

- SUPER undergraduate research program	2026
- Aresty science research program	2026 - present
- NSF REU DIMACS program	2026
- RAI4Ukraine research program	2024-2025

Organizer/Chair:

- ASA SLDS Section "The role of the Rashomon Effect in Responsible AI"	2026
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- ASA SLDS Section “Transparent Machine Learning through Visualization and Human-Centered Design” 2026
- INFORMS Annual Meeting session “Emerging Trends in Interpretable Machine Learning” 2024
- Discussion series “Controversial Topics in Precision Medicine and Learning” (collaboration between SAMSI and Duke Computer Science) 2019

Area Chair:

NeurIPS: 2025, 2026

ICLR: 2027

Reviewer:

NeurIPS: 2024, 2025

ICLR: 2025

AISTATS: 2025

ICML: 2025, 2026

KDD: 2025

CVPR: 2026

ACL ARR: 2026

Journals: JMLR, Patterns, ACM Journal on Responsible Computing

Workshops: DeepMath (2022, 2023), International Workshop on Advances in Interpretable Machine Learning and Artificial Intelligence (2021, 2022, 2023), Workshop on Principled Design for Trustworthy AI - Interpretability, Robustness, and Safety across Modalities @ ICLR 2026, Algorithmic Fairness Across Alignment Procedures and Agentic Systems @ ICLR 2026

Panelist:

Microsoft Research Undergraduate Internship 2024

Alumni Board Member of Duke Friends of Ukraine 2024 – present

Volunteer:

- Procurement manager at “Razom for Ukraine” nonprofit 2022-2023

- KDD 2017

- ICML 2019

- Study Zone tutor at King County Library System (provided homework help to K-12 students) 2015

Student Assistant:

- NSF-sponsored Seamless/Seamful Human Technology Interaction (HTI) Workshop May 2021

- Graduate student committee member for the Faculty Search and Prospective Student Visits for the Department of Computer Science, Duke University 2016–2019

Alumni:

- ComSciCon Triangle (science communication workshop for graduate students) 2018

Head of Scientific Association of Students and Postgraduates of Taras Shevchenko National University of Kyiv 2012 – 2013

SERVICE
at RUTGERS

Faculty advisor

- RAISE ([Rutgers AI Safety Exchange](#)) Spring 2026 – present

- Rutgers Artificial Intelligence in Medicine and Healthcare Association Fall 2026 – present

Judge at Rutgers Hackaton HackRU Fall 2025

CS Department PhD admission committee 2026, 2027

Committee member for Qualification Exam:

- for Rutgers CS PhD Student Zihe Ye 2025

- for Rutgers CS PhD Student Yash Sharma 2026

PhD Thesis Committee:

- for Rutgers CS PhD Student Hengyi Wang
- for Rutgers ECE PhD Student Sinjini Banerjee

2026

2026